ASSIGNMENT #1

Course:	**CHE 433. PROCESS SIMULATION WITH ASPEN PLUS**
Term:	**Fall 2019**
Professor:	**Dr. Michael Caracotsios**
Office:	**EIB 244**
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Your name:____

Due Date: Friday September 06 by 5:00pm in my office

Number of points...........................100

Your score................................

Note: **All problems are worth 100 points. You final score will be the average value**

Problem 1: **Review of Macroscopic Balances and Numerical Methods Used in Commercial Process Simulators.**

A fluid is flowing in a linear($L = 36\ ft$) standard pipe $Schedule\ 40,\ 1in$ with a dirt factor of $k = 0.0077\,mm$. The Reynolds number is $\mathrm{Re} = 318,000$. Using the Colebrook equation calculate the friction factor using the methods listed below:

1. Solve by the Newton method to obtain the friction factor.
2. Repeat by using the Secant Method
3. Repeat by using the Direct Substitution method
4. Repeat by using the Broyden method
5. Repeat using the Wegstein acceleration method. Perform two passes before accelerating
6. Calculate the pressure drop along the linear pipe if the fluid is air that enters the pipe at $\left\{P_1 = 20\,psig, T_1 = 70\,^oF\right\}$ and leaves the pipe at $T_2 = 170\,^oF$. What is the rate of heat transfer to the air.
7. **[Optional]** Create an ASPEN Plus simulation to verify your result in Part 6.

Notes: For your numerical computations you may use EXCEL or perform hand calculations. In all cases, clearly state the algorithmic formula that you are using and prepare tables of the iterations sequence with a clear indication of your convergence criterion. If any of the above methods is not converging to the solution indicate the reason why